Journée des Jeunes Chercheurs en Audition, Acoustique musicale et Signal audio

26 et 27 Janvier 2026, Le Mans

ANOMALY DETECTION USING EXTRACTED ORDERS AND AUTO ENCODERS

Anouck Bruguière[a],[b], Kais Hassan[a], Jean-Hugh Thomas[a], Amaury Graillat[b]

[a] **Laboratoire d'Acoustique de l'Université du Mans (LAUM), UMR 6613,** *Institut d'Acoustique - Graduate School (IA-GS), CNRS, Le Mans Université, France*

[b] **Oros Digital, Digigram Group,** *Grenoble, France*

anouck.bruguiere@oros.com

1 CONTEXT

Industrial data is complex, predominantly **healthy**, and unlabeled, making **traditional** supervised learning **unsuitable** for mitigating the high costs of unplanned shutdowns.

- **Support Vector Machine (SVM)** are effective, but are **limited** by the need for manually defined features[Liu18, Lei18]
- **Deep Learning (DL)** can automatically extract features for end-to-end diagnosis, but often lacks **explainability** and requires extensive labeled datasets that are rarely available in real-world industrial settings [Cui21]

OBJECTIVE

Avoid unplanned shutdowns of rotating machinery by detecting shifts from the nominal behavior using healthy data and industrial validation

2 KEY CHALLENGES

- Explainable & Interpretable AI (XAI)*
- Handling unlabeled & Imbalanced data
- Robustness & Adaptation in evolving environments

**Explainable Artificial Intelligence*

3 FEATURE-EXTRACTION STRATEGIES

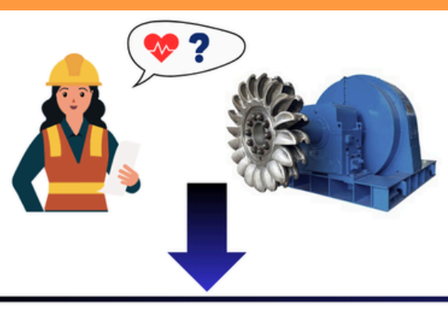
Power Spectral Density estimation
Welch's Method: Overlapping segments → Modified periodograms → Averaging (PSD estimate)

Harmonic targeting
(Orders X1, X2, X3 & X4)

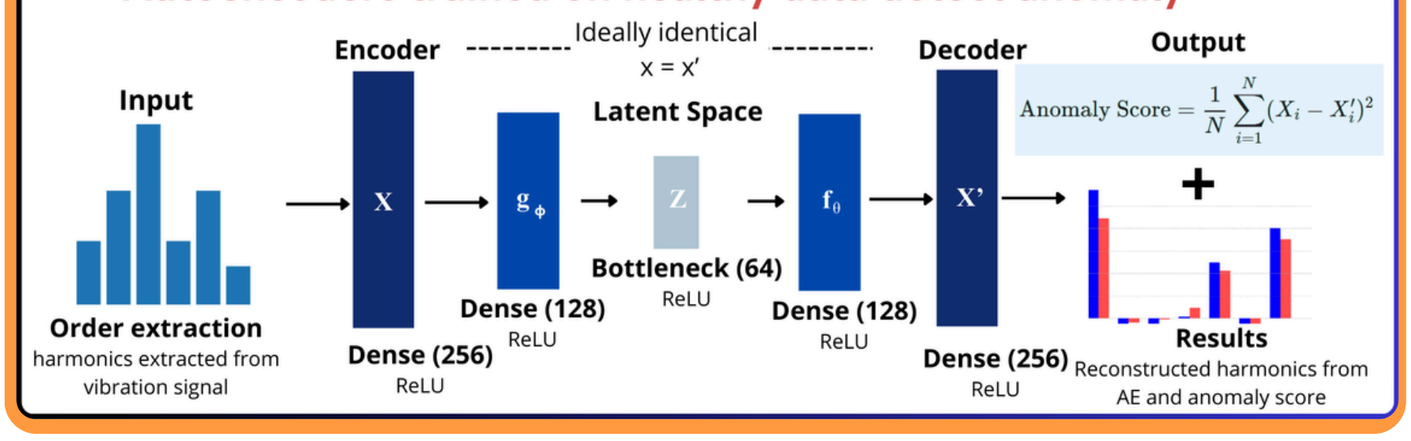
Amplitude extraction
via Interpolation

Feature vector construction
10 Harmonics x 4 Accelerometer

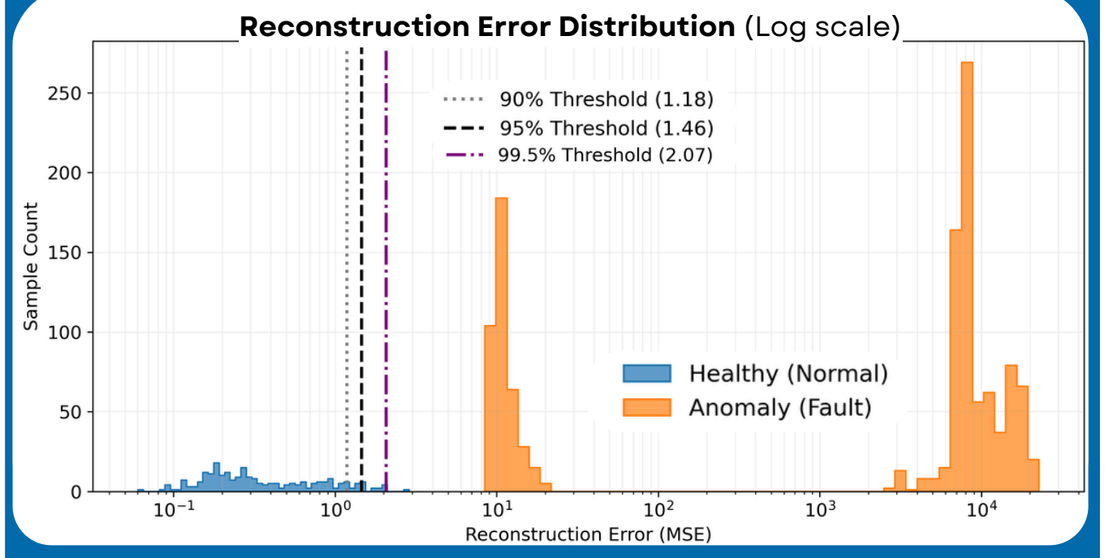
4 METHODOLOGY

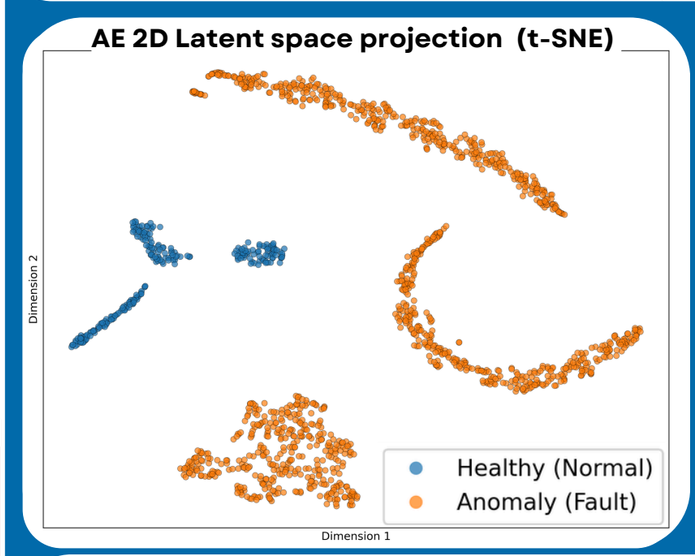


Autoencoders trained on healthy data detect anomaly



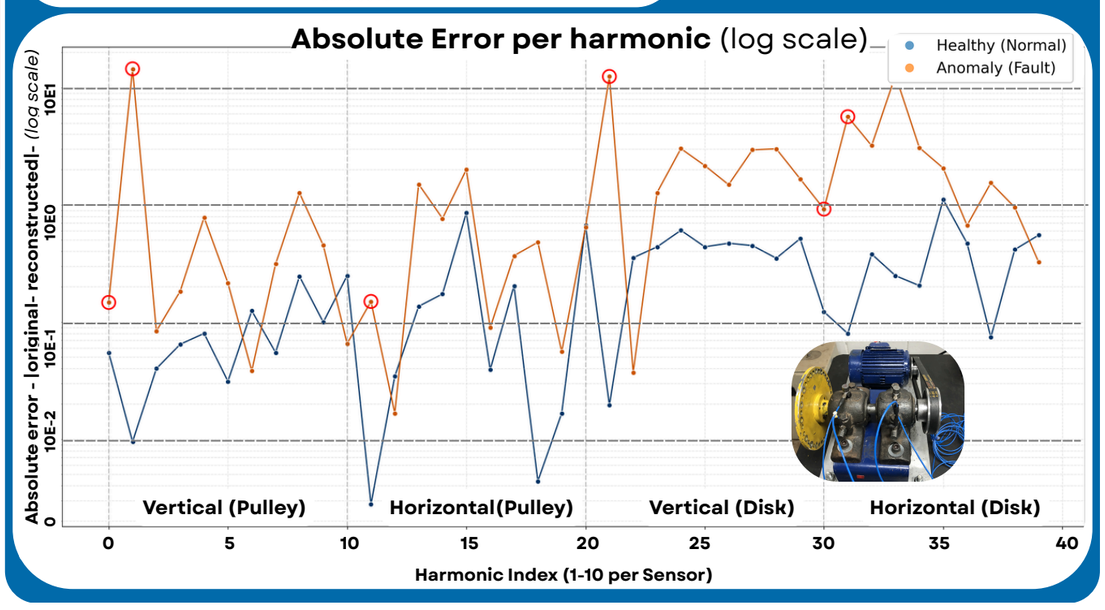
5 RESULTS ON MFRM▲ DATASET





Metrics	Value [%]
Accuracy	99.86
F1-Score	99.92
Precision	99.83
Recall	100.00

▲ *Mechanical faults in rotating machinery, DOI: 10.17632/zx8pfhdtnb.3*



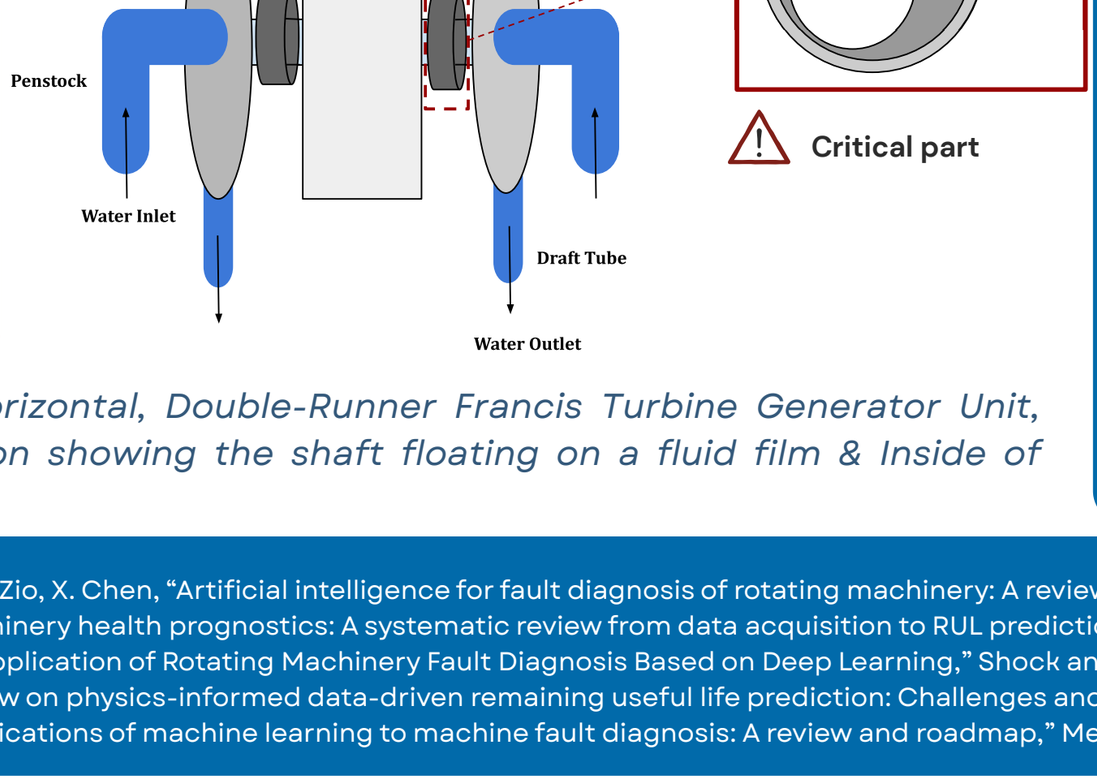
6 NEXT STEPS: FRANCIS TURBINE VALIDATION

Target Faults

Unbalanced rotor
&
Shaft Misalignment

Measured data

- Acceleration signal of casing 1&2, and Plain bearing 1&2
- Axial Displacement
- Rotor's velocity



Schematic of Horizontal, Double-Runner Francis Turbine Generator Unit, with cross-section showing the shaft floating on a fluid film & Inside of Francis turbine

Related literature

[Liu18] R. Liu, B. Yang, E. Zio, X. Chen, "Artificial intelligence for fault diagnosis of rotating machinery: A review," Mechanical Systems and Signal Processing, vol. 108, pp. 33-47, 2018.
[Lei18] Y. Lei, et al."Machinery health prognostics: A systematic review from data acquisition to RUL prediction," Mechanical Systems and Signal Processing, vol. 104, pp. 799-834, 2018.
[Cui21] W. Cui, et al. "Application of Rotating Machinery Fault Diagnosis Based on Deep Learning," Shock and Vibration, vol. 2021, 3083190, 30 pages, 2021.
[Li24] H. Li, et al "A review on physics-informed data-driven remaining useful life prediction: Challenges and opportunities," Mechanical Sytems and Signal Processing, vol. 209, 111120, 2024.
[Lei20] Y. Lei. et al "Applications of machine learning to machine fault diagnosis: A review and roadmap," Mechanical Systems and Signal Processing, vol. 138, 106587, 2020.